Fiscal Year:	FY 2011	**Task Last Updated:**	FY 01/26/2011
PI Name:	Durian, Douglas J. Ph.D.		
Project Title:	The Melting of Aqueous Foams		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	COMPLEX FLUIDS/SOFT MATTER--Complex Fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	19104-6396	**Congressional District:**	2
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	1995 Microgravity Fluid Physics NRA-94-OLMSA-05: Research and Flight Experiment Opportunities
Start Date:	08/05/2004	**End Date:**	04/04/2013
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:	NOTE: Grant number was changed in 2007 to NNX07AP20G (Ed., 1/4/13) NOTE: Funding extended to 4/04/2013 per NSSC database (5/2010) NOTE: Funding extended to 4/4/2010 through a new grant in 2007 (6/2008) NOTE: Funding reinstated thru 9/30/2008, per F. Chiaramonte (8/2006). Suppl funding thru Oct 2006 (6-5-06)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX07AP20G ; NNX07AK60G ; NNC04GB61G		
Performance Goal No.:			
Performance Goal Text:			

The unusual elastic character of foams will be quantified macroscopically by measurement of the shear stress as a function of static shear strain, shear strain rate, and time following a step strain; such data will be analyzed in terms of a yield stress, a static shear modulus, and dynamical time scales. Microscopic information about bubble packing and rearrangement dynamics, from which these macroscopic non-Newtonian properties presumably arise, will be obtained non-invasively by novel multiple-light scattering diagnostics such as diffusing-wave spectroscopy (DWS) and speckle-visibility spectroscopy (SVS). Bubble size and homogeneity will be studied by diffusely transmitted light (DTS). Liquid fraction will be measured by electrical conductivity. Quantitative trends with materials parameters, such as average bubble size, and liquid content, will be sought in order to elucidate the fundamental connection between the microscopic structure and dynamics and the macroscopic rheology.

There are three all-encompassing objectives motivating the FOAM project. The first concerns multiphase flows and represents the

Task Description:	"mechanics" aspect alluded to in the F.O.A.M. acronym: To understand how foams and bubbly liquids deform and flow in response to applied forces. The second objective concerns multiphase separations: To understand how foams and bubbly liquids evolve with time via a combination of liquid flow between bubbles, of gas diffusion across the film separating neighboring bubbles, and of direct coalescence via film rupture. For both flows and separations, some level of understanding has been established for very dry foams, and also for liquids with relatively few bubbles; however, we are more ignorant of the crucial intermediate regime of very wet foams and very bubbly liquids. This regime defies earth-bound experiments. The third, overall, objective is to transfer this new understanding of multiphase flows and separations into ASL technology for NASA's strategic exploration mission. The utility and fascination of foams are derived largely from the surprising fact that they have a solid-like elastic character in spite of being mostly gas with a few percent volume fraction of liquid, but can nevertheless flow under shear. The physical origin of such unusual rheology in terms of microscopic structure and dynamics is poorly understood and remains a subject of basic scientific interest to physicists, chemists, and chemical engineers. The proposed research promises important new insight into these issues, and could also have significant consequences for our understanding of flow in other dense randomly-packed systems such as emulsions, colloidal suspensions, slurries, bubbly liquids, and granular materials. Furthermore, all foam applications are empirically based and the proposed research may generate valuable fundamental guidance for the development of materials with more desirable rheology and better stability. The value of the FOAM project in terms of NASA's exploration mission lies with Advanced Life Support. Water recovery and recycling are absolutely central to ASL, and require fine control of multiphase flows and separations. The very heart of the FOAM project is to bring new technology and understanding to these problems
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Foams are important for a clean environment. Firstly, their presence in natural waters signals the presence of surface-active impurities and a need for clean-up. Secondly, they are useful themselves in the cleanup process since many impurities can be removed by injecting gas and skimming off the resulting foam. Relatedly, study of foams brings specific insight into multiphase flow and separation phenomena of direct importance for water recovery and recycling operations.
Task Progress:	During this period my group included Penn graduate student Adam Roth and visiting summer intern Victor Sebag. During the past year we have made significant progress on several different aspects of the overall FOAM project: First I helped finalize the new ESR (Experimental Requirements Document) and interacted with the ESA engineering team. And I spoke at the FOAM-C Utilization meeting at ESA headquarters in Paris. Due to both financial and technical contraints, the focus in the near term has shifted to coarsening. This is not a new direction, but rather a subset of the originally envisioned and funded ultimate microgravity experiments. In addition to imaging of surface bubbles, the planned restricted experiment features optical probes, like DTS, DWS, and now SVS, as crucial diagnostics of the structure and dynamics. In terms of communicating results, I gave invited presentations on foams at the Institute for Advanced Studies in Princeton and at the APS March Meeting in Dallas, plus my student Adam Roth gave two contributed talks and one poster presentation. Furthermore I published two peer-reviewed review articles on foams, one in Physics Today and one in a book; plus my students and I have completed rough drafts for two research articles. Research accomplishments all pertain to the structure and the coarsening of 2d and confined foams. The most major effort involved use of a novel cell in which bubbles are confined between parallel plates and the liquid content is varied by a hydrostatic pressure set by the filling depth of a trough that surrounds the sample. Here we collected time sequences of digital images, and wrote software to extract bubble areas and side numbers. We discovered that the distributions of these quantities are stationary, and independent of liquid fraction. And we discovered that the von Neumann law for the rate of coarsening of ideal dry 2d foams has limitations for small bubbles due to nonzero liquid fraction effects. We succeeded in modeling the data by a Plateau border blocking model, where the diffusion of gas between neighboring bubbles is only through the films and not at all through the borders. While this captured the behavior for the rate of change of bubble areas, there is a fitting parameter that represents the size of the borders and the results for this quantity are only is fair agreement with expectation based on equality of Laplace and hydrostatic pressure differences as needed for mechanical equilibrium. We suspect that gas actually does diffuse across the Plateau borders. This work is complete, and is being written up for publication. In addition to varying the wetness of 2d foams, we also embarked on an experiment where the geometry of the embedding space is curved. This alters the von Neumann law in a way predicted some time ago but never tested in experiment. We currently have constructed a sample cell, where the foam is confined between concentric clear Lucite domes. And we have developed an image processing procedure to eliminate the "double image" of the borders of the bubbles which are away from the north pole, at the top of the dome, where the camera is located. This work is well underway and we hope to complete it during the coming year. We also have conducted experiments on the final states of foams that coarsen in a confined geometry. The primary example is the so-called "bamboo" state when the foam is in a tube and all bubbles become cylinders separated by circular flat films perpendicular to the tube axis. Here we discovered that the length distribution is independent of the liquid filling fraction. And we accumulated statistics 1000x better than those of Fortes. The distribution we find may be modeled simply. This work is now complete, and is being written up for publication. A related experiment is now underway, in which a 2d foam coarsens to a final state inside a flat cell with a hexagonal boundary. This work should come to completion during the coming year. While the above on-going experiments and papers are being brought to completion, we will begin a new series of experiments on the evolution and shape statistics of edge bubbles of dry 3d foams. We wish to discover if the distributions are stationary, and if they depend upon liquid content. On earth the liquid content can be varied only minimally when homogeneous. Large gradients and correlations with bubble size necessarily occur if the foam is very wet; hence the need for extended microgravity conditions in the upcoming flight experiments.
Bibliography Type:	Description: (Last Updated: 06/12/2025)
Abstracts for Journals and Proceedings	Durian DJ. "Making a frothy shampoo or beer." Invited presentation. March 2011 Meeting of the American Physical Society, Dallas, Texas, March 21-25, 2011. Bulletin of the American Physical Society 2011;56(2):V5.00005. , Mar-2011
Articles in Peer-reviewed Journals	Durian DJ, Raghavan SR. "Making a frothy shampoo or beer." Physics Today 2010 May;63(5):62-3. http://ptonline.aip.org/getpdf/servlet/GetPDFServlet?filetype=pdf&id=PHTOAD000063000005000062000001&idtype=cvips&bypassSSO=1 , May-2010
Books/Book Chapters	Dauchot O, Durian DJ, van Hecke M. "Dynamical heterogeneities in grains and foams." in "Dynamical heterogeneities in glasses, colloids, and granular media." Ed. L. Berthier, G. Biroli, J.-P. Bouchoud, L. Cipelletti, and W. van Saaroos. Oxford University Press, submitted October 2010. To be published in 2011. http://arxiv.org/abs/1010.0873 , Oct-2010